Karthik Kesavarapu

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Summary

M.S. CS (AI) student at UChicago with an interest in Data Engineering, AI, and Full-Stack Development. Built Medallion pipelines on Azure Databricks and Unity Catalog and automated workflows using multi-agent architecture that increased productivity.

Education

University of Chicago Sept 2026 – Mar 2028 (Expected)
Master of Science in Computer Science, Artificial Intelligence Specialization *Chicago, IL*

University of Illinois Chicago (UIC) Aug 2023 – May 2026
Bachelor of Science in Computer Science, College of Engineering *Chicago, IL*

- GPA: 3.9/4.0 | Coursework: Machine Learning, Database Systems, Agentic AI

Technical Skills

Languages: Python, SQL, TypeScript, JavaScript, Java, C++, C, Bash
Data Engineering: Azure Databricks, Unity Catalog, Medallion Architecture, ETL Pipelines, Data Validation
AI/ML: LangGraph, LangChain, OpenAI API, RAG, ChromaDB, Pydantic, scikit-learn, Pandas
Databases: PostgreSQL, SQL Server, MySQL, SQLite
Frameworks: FastAPI, Flask, Next.js, React, Node.js, Streamlit
Cloud & Tools: Microsoft Azure, AWS (EC2), Power BI (DAX), Git, Azure DevOps, JIRA
Certifications: Databricks Certified Data Analyst Associate; Databricks Academy: AI Agent & Generative AI Fundamentals

Experience

Old National Bank May 2026 – Aug 2026
Treasury Management Analyst Intern — Azure Databricks, Power BI *Chicago, IL*

- Built a centralized data-validation report on Databricks queries to monitor dataset health across all bank domains, checking load status, row counts, and budget values.
- Designed configurable validation thresholds giving data engineers per-dataset control over check strictness.
- Redesigned treasury KPI reports and optimized DAX measures to reduce Power BI load times.

Seyon Solutions Aug 2025 – Jan 2026
Data Engineering Intern — Azure Databricks *Chicago, IL*

- Implemented Medallion Architecture in Azure Databricks to ingest, clean, and transform data across Bronze, Silver, and Gold layers governed by Unity Catalog.
- Built and optimized ETL pipelines to validate and enrich raw data for analytics readiness.
- Designed Gold-layer datasets optimized for business reporting and downstream analytics.

Seyon Solutions May 2025 – July 2025
Data Engineering Intern — Power BI Developer *Chicago, IL*

- Developed Power BI dashboards integrating SQL, Excel, and cloud sources, modeling data with Power Query and DAX and enabling natural-language querying via Q&A.

Real AI Dynamics Aug 2024 – Dec 2024
Next.js Developer

- Built real-time communication components in Next.js and integrated backend APIs with the UI in Agile sprints using GitHub and JIRA.
- Refactored components into modular structures to improve front-end performance.

Projects

TM Next Product | *Python, LangGraph, OpenAI API, FastAPI, Next.js, ChromaDB*

- Built a proof-of-concept copilot for Old National Bank's Treasury Management consultants: a six-stage LangGraph pipeline (profile, needs, product, ROI, compliance, executive summary) with Pydantic-typed agent outputs.
- Retrieved candidate products from a ChromaDB knowledge base via semantic search, paired with a deterministic ROI engine and a consultant accept/reject feedback loop that informs future recommendations.
- Streamed agent progress live to a Next.js dashboard via Server-Sent Events from an async FastAPI backend, persisting analyses in SQLite.

Educational Chatbot | *Next.js, TypeScript, FastAPI, Python, ChromaDB, OpenAI API* GitHub

- Architected a multi-agent tutor whose Socratic agent guides students through adaptive questioning, not answers.
- Built a RAG pipeline on a ChromaDB vector store to ground responses in course-specific material.
- Designed FastAPI REST endpoints for the Next.js frontend, with JWT authentication and a lockdown exam mode.

XOPYops | *Python, PostgreSQL, scikit-learn*

- Built a full-stack industrial operations app simulating 6 months of machine health, applying logistic regression and random forest models for downtime risk analysis.